

a dangerous level of radiation and the buzzer will start beeping continuously.

What the different LED colours mean

Indication of measured microwave power in the frequency range of 800 MHz to 4 GHz by different LED colours has been decided after interacting with hundreds of people where measurements have been carried out, and interviewing them about possible health hazards experienced by them or their family members and colleagues, Prof. Kumar reports. For radiations below -30 dBm, most people report no health problems, so this is indicated by the green LED. However, there are reports that some electro-sensitive individuals do experience adverse health effects even at radiation levels as low as -40 dBm to -50 dBm.

Between -30 dBm and -15 dBm, some individuals complain of headaches, sleep disturbances, joint pain, memory loss, stress, fatigue, skin rashes and even miscarriages over a few years of exposure, so this is indicated by a yellow LED. Above -15 dBm, many individuals have reported the above-mentioned health problems within a year of exposure and a few cases of cancer have been observed within a few years; so this is indicated by the red LED. The above health effects are based on individual sensitivity or immunity.

What to do if radiation levels are high

If cell-tower radiation is high at your premises, which is indicated by the red LED and/or the yellow LED, you can request your government or telecom regulatory body to adopt stricter radiation norms. You can also request cell operators to reduce the transmitted power, ask them to change the direction of the transmitting antenna or even remove the tower. It is possible that the above options may take time to get implemented. Meanwhile, you can take the following precautions.

Features

Regulatory Model No.: Detex 189

Typical microwave detector and internal antenna frequency response: 800 MHz to 4000 MHz

Radiation power measurement range: -50 dBm to +10 dBm (power density of 1,000,000 micro watts per square metre)

LED scale points:

Only green LED 'on': Power level less than -30 dBm ± 3 dBm over the frequency range.

Yellow and green LEDs 'on': Power level between -30 dBm and -15 dBm with an accuracy of ± 3 dBm over the frequency range.

Red, yellow and green LEDs 'on': Power level greater than -15 dBm ± 3 dBm over the frequency range

Power source: Li-ion rechargeable battery (3.7V, 1000mAh) {Part No.: 189-B or any model equivalent to Nokia BL-5C}

Maximum current consumption: 50 mA at 3.7 V when all LEDs are 'on' and the buzzer is beeping

Rechargeable battery life: Approximately 2 to 3 years.

Operating time: Once fully charged, the device can work up to 24 hours, depending on the mode of operation.

Device dimension: 200 $\times$ 80 $\times$ 25 mm (at the top) and $\times$ 42 mm (at the bottom)

Package size: 250 $\times$ 120 $\times$ 60 mm

Device weight: 250 grams, including battery

Package weight: 350 grams, including battery

Operating temperature: 0°C to 50°C

Humidity: 90 per cent RH, non-condensing

Challenges faced in development of this product

There were several challenges faced while developing Detex 189. A few are listed below:

1. The design of the planar broadband elliptical monopole antenna to cover the frequency range of 800 GHz to 4 GHz.
2. The design of the broadband impedance matching network with an input impedance of Schottky diodes.
3. The design of the voltage-doubler circuit.
4. The decision to set various threshold levels to display safe, caution and danger levels, based on health hazards.
5. To arrive at a compact and ergonomically pleasing design without affecting core functionality.

1. Avoid places where there is high radiation, especially when the red LED is 'on.'

2. Try to spend as little time as possible where radiation is high, indicated by the red and/or yellow LED.

3. You can 'shield' your place, if the radiation level is high. These 'shields' reduce the radiation levels considerably, which can be confirmed with Detex 189. NESA also provides shielding solutions in the form of cellphone tower-radiation-absorbing fabric curtains, wall papers, window film glass, personal shields and pregnancy aprons. The different kind of shields might sound ordinary but they are integrated with built-in CML (capacitive matched load) electronic chips to absorb the radiation. A reduction in

radiation of more than 99 per cent for curtains and wallpapers, and about 95 to 97 per cent for window film glass, is claimed by NESA.

In the future

"We are planning to add more features to it, such as a digital display and more LED displays," says Prof. Kumar. He adds, "We are in the process of filing for patents too." The product has been designed and tested at Antenna Lab, IIT Bombay. The technology has been transferred for mass manufacturing to NESA Radiation Solutions Pvt Ltd, and the product will be sold exclusively by it. The company has already sold a number of units. ●

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